Week 4: Data Visualization & Capstone Project

End-to-End Business Performance Analysis

Dataset: Sample data (1).xlsx | 700 records | 16 source fields

Metric	Result
Total Sales	118,726,350.26
Total Profit	16,893,702.26
Profit Margin	14.23%
Units Sold	1,125,806.00
Gross Sales	127,931,598.50
Discounts	9,205,248.24

Executive summary

The dataset contains 700 records spanning 2013–2014. Total sales are 118,726,350.26 and total profit is 16,893,702.26, producing an overall profit margin of 14.23%. United States of America is the leading country by sales, Paseo is the leading product by sales, and Government is the leading segment by sales.

Data preparation and quality

Column names were trimmed for consistency, the Date field was parsed as a date, and the 53 missing Discount Band values were labeled 'Not Specified' rather than guessed. No exact duplicate rows were detected. No other missing values were found. All 700 records were retained.

Key findings

• Country leader: United States of America with sales of 25,029,830.16.

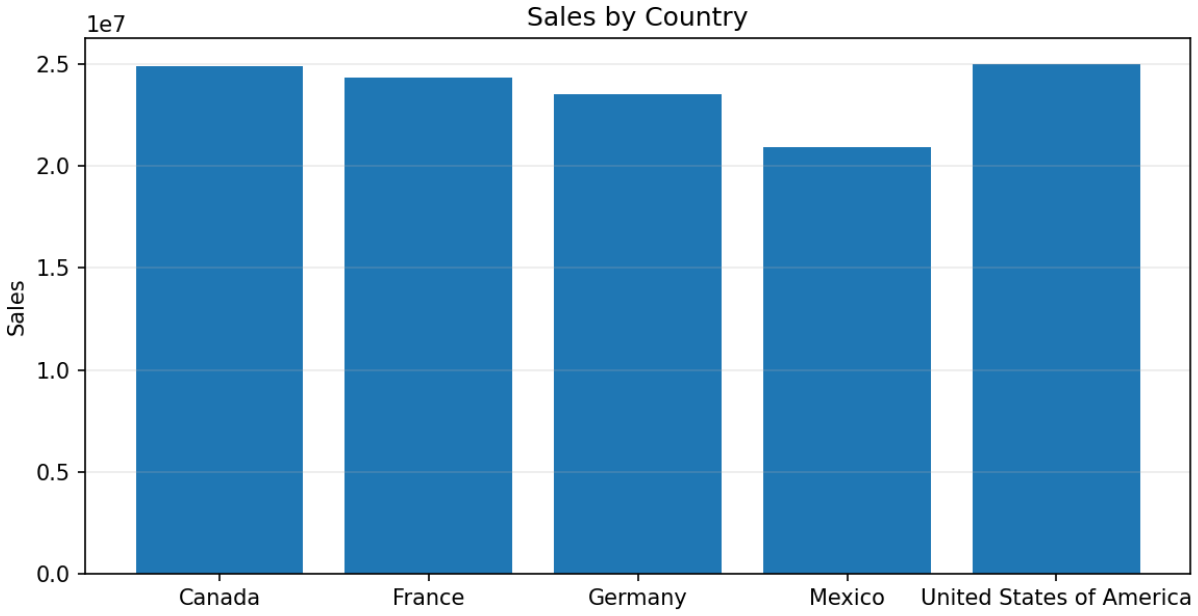
• Product leader: Paseo with sales of 33,011,143.95.

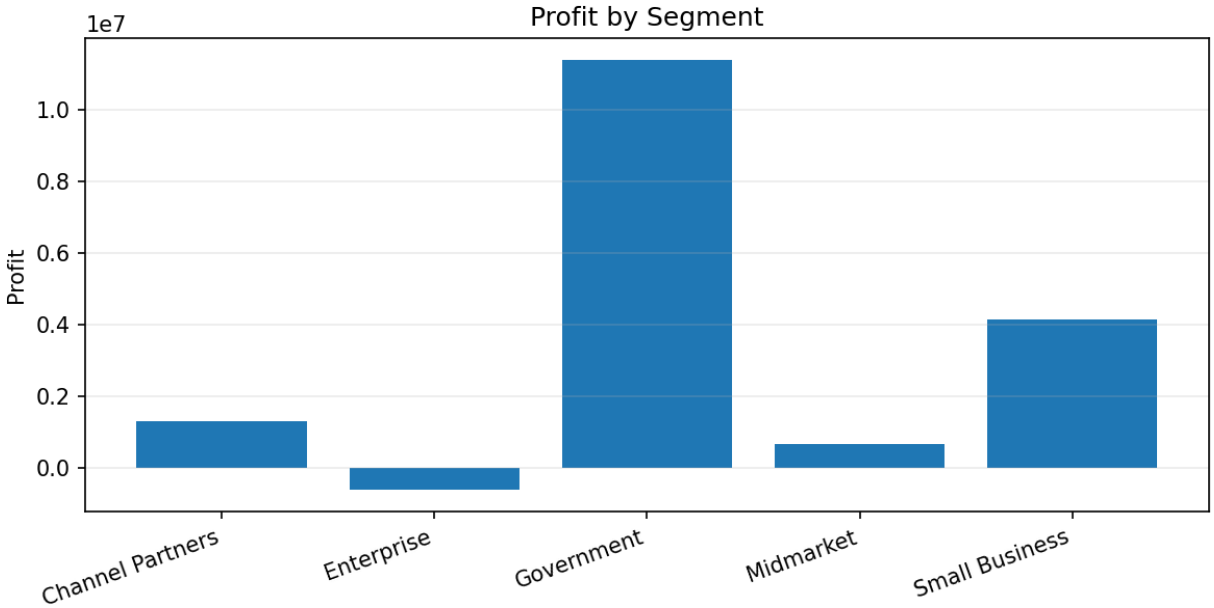
• Segment leader: Government with sales of 52,504,260.67.

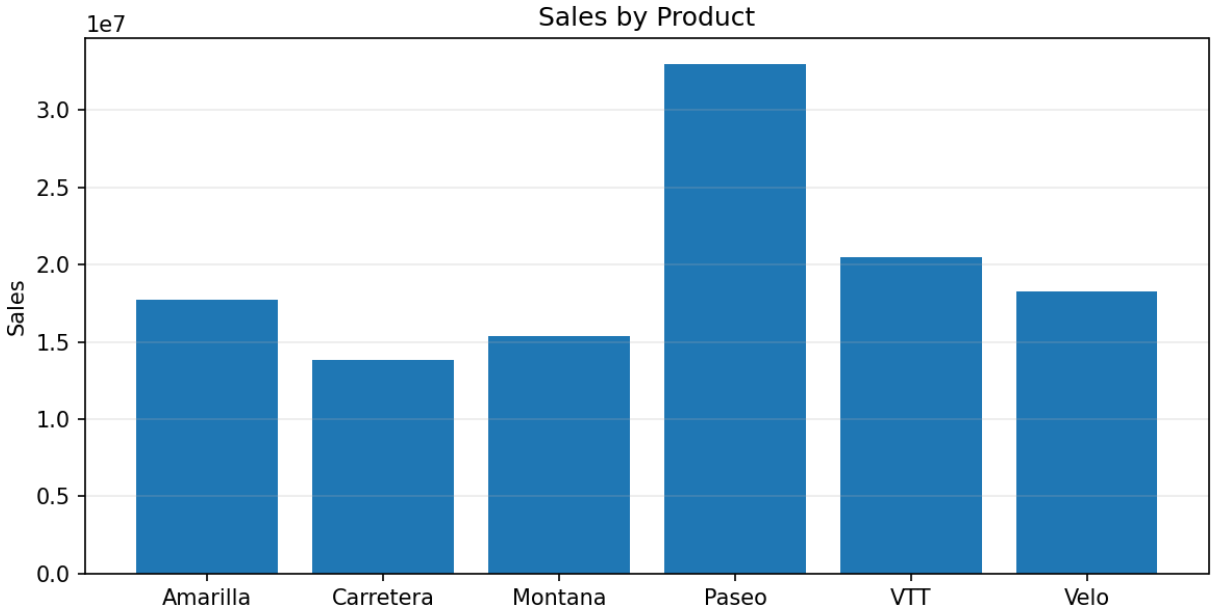
• Highest-sales month: October 2014 (12,375,819.92); lowest-sales month: September 2013 (4,484,000.03).

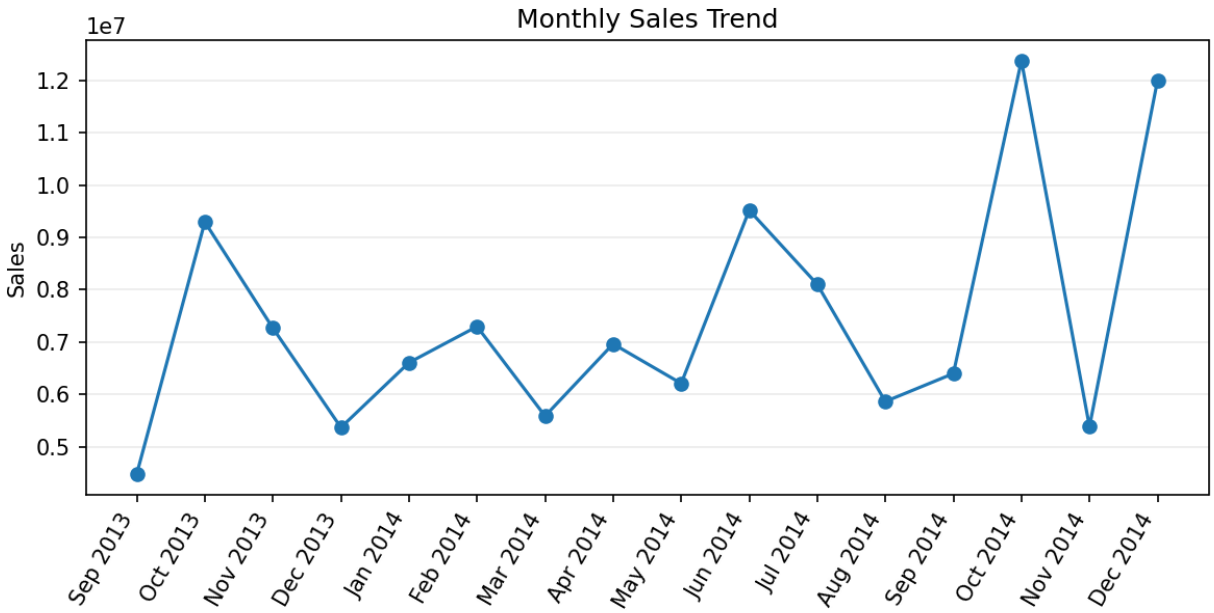
• 58 records have negative profit and should be investigated for pricing, discounting, or cost drivers.

Visual analysis


Sales by Country
1e7
2.5
2.0
1.5
1.0
0.5
0.0
Sales
Canada
France
Germany
Mexico
United States of America


Profit by Segment
1e7
1.0
0.8
0.6
0.4
0.2
0.0
Profit
Channel Partners
Enterprise
Government
Midmarket
Small Business





Recommendations

• Prioritize country, product, and segment combinations with strong sales and profit; use the workbook summary tabs to identify where profit differs from revenue.

• Review all negative-profit records and check their discount, sale price, manufacturing price, and COGS before changing pricing policy.

• Compare monthly sales and profit together in a BI dashboard to distinguish high-revenue periods from high-profit periods.

• Treat 'Not Specified' discount-band records as a data-quality follow-up and confirm whether the source should contain those labels.

Dashboard / BI implementation guide

Use the Cleaned Data tab as the Power BI or Tableau source. Recommended KPI cards: Sales, Profit, Profit Margin, Units Sold and Discounts. Recommended visuals: sales by country, profit by segment, sales by product, monthly sales/profit trend, and discount-band performance. Add slicers for Year, Country, Segment, Product and

Discount Band.

Note: The supplied Week 4 PDF asks for a Power BI or Tableau dashboard, but neither application project file can be generated here. The Excel workbook includes a dashboard-style summary and charts, plus cleaned data and grouped tables suitable for importing into Power BI/Tableau.